



BIKANER TECHNICAL UNIVERSITY, BIKANER
बीकानेर तकनीकी विश्वविद्यालय, बीकानेर
OFFICE OF THE DEAN ACADEMICS



VI Semester	
B. Tech. (Artificial Intelligence & Data Science)	
6AD4-22: Statistical Modeling and Forecasting Lab	
Credit: 1	Max. Marks: 100 (IA:60, ETE:40)
0L+0T+ 2P	End Term Exams: 2 Hours
Course Objectives: As a result of successfully completing this course, students will : • Able to understand basic property of time-series data. • Able to handle seasonality and trend in time series data. • Able to use and deploy various models for time series data. • Able to select best model for time series data.	
Course Outcomes: Upon successful completion of the course, students will be able to CO-1: Discuss the challenges and their solutions for Time Series Data. CO-2: Understanding impact of seasonality in time series data. CO-3: Understand role of drift and trend. CO-4: Understand to working of various models used for time series data.	
S. No.	List of Experiments
1	Cleaning, Preprocessing and Handling Time Series Data • Time Series Data Cleaning • Loading and Handling Times series data Preprocessing Techniques
2	How to Checking Stationarity of a Time Series and making Time Series data Stationary Estimating & Eliminating Trend. • Aggregation • Smoothing • Polynomial Fitting Eliminating Trend and Seasonality • Differencing Decomposition
3	Time Series analysis a) Moving Average time analysis data. b) Smoothing the Time analysis Data. Check out the Time series Linear and non-linear trends.
4	Time Series Modelling and Forecasting • Moving average • Exponential smoothing • ARIMA Seasonal autoregressive integrated moving average model (SARIMA)
5	Dependence Techniques • Multivariate Analysis of Variance and Covariance • Canonical Correlation Analysis • Structural Equation Modeling Inter-Dependence Techniques • Factor Analysis

Approved by academic council meeting held on

Office: Bikaner Technical University, Bikaner

Karni Industrial Area, Pugal Road, Bikaner-334004; Website: <https://btu.ac.in>

Page 39



BIKANER TECHNICAL UNIVERSITY, BIKANER
बीकानेर तकनीकी विश्वविद्यालय, बीकानेर
OFFICE OF THE DEAN ACADEMICS



	Cluster Analysis
Suggested Books: 1. Introduction To Time Series Analysis And Forecasting, 2nd Edition, Wiley Series In Probability And Statistics, By Douglas C. Montgomery, Cheryl L. Jen(2015) 2. Master Time Series Data Processing, Visualization, And Modeling Using Python Dr. Avishek Pal Dr. Pks Prakash (2017) 3. Chris Brooks “Introductory Econometrics for Finance,” Fourth Edition, Cambridge University Press 2019 4. Ruey S. Tsay “Analysis of Time-series data,” Third Edition, Wiley 2014 5. John Fox and Sanford Weisberg “An R Companion to Applied Regression,” Third Edition, SAGE 2018 6. Yves Croissant and Giovanni Millo “Panel Data Econometrics with R,” First Edition, Wiley 2018	